

19. (3.25 points)

An actuary is reviewing results of a hurricane model on a potential new reinsurance program. The actuary simulates five years of losses with the following results:

Combined Ratios			
Simulation	Direct	Net of Current Program	Net of Proposed Program
#1	130%	105%	115%
#2	180%	140%	130%
#3	80%	90%	100%
#4	90%	95%	105%
#5	200%	160%	150%

a. (1.25 points)

Construct a coverage option cumulative distribution function chart using the simulations above.

b. (0.5 point)

Select a reinsurance program and briefly explain your selection based on the simulations above if an actuary is primarily concerned about losses in the tail.

c. (0.75 point)

Calculate the 80th percentile results using direct data for the following:

- i. $V@R$
- ii. $TV@R$
- iii. $XTV@R$

d. (0.25 point)

Define $WTV@R$.

e. (0.5 point)

Briefly describe a weakness of $TV@R$ and briefly explain how $WTV@R$ corrects the weakness.

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SAMPLE ANSWERS AND EXAMINER'S REPORT – SPRING 2019 EXAM 7

SPRING 2019 EXAM 7, QUESTION 19

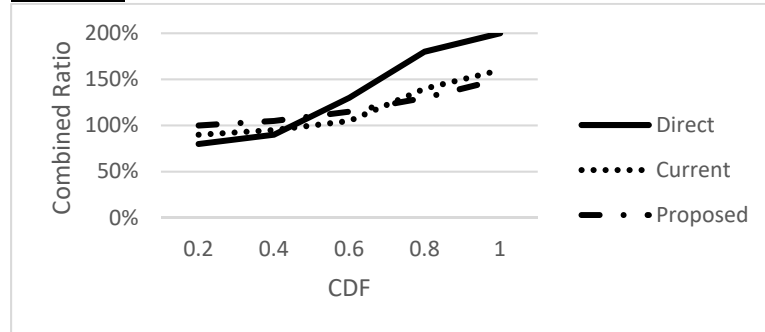
TOTAL POINT VALUE: 3.25

LEARNING OBJECTIVE(S): C2, C3

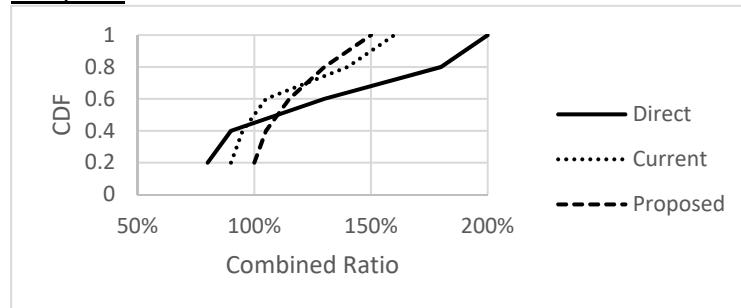
SAMPLE ANSWERS

Part a: 1.25 points

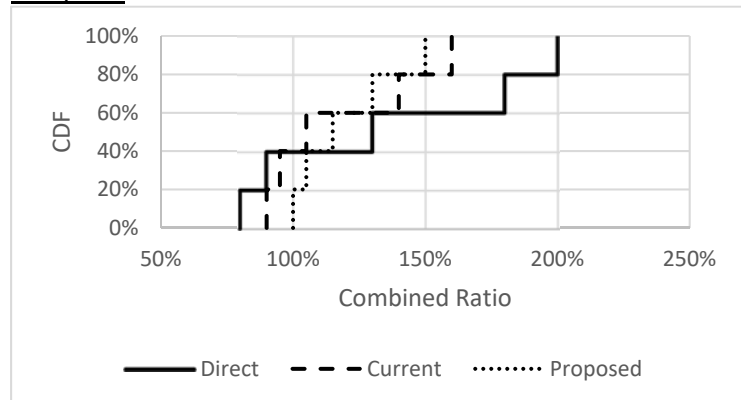
Sample 1



Sample 2



Sample 3



Part b: 0.5 point

Sample 1

Proposed program has lower Combined Ratio in the tail, use proposed.

Sample 2

I choose proposed because it outperforms current & bare at around $\alpha = .7$ and above, even though it performs worse at lower α 's.

SAMPLE ANSWERS AND EXAMINER'S REPORT – SPRING 2019 EXAM 7

Part c: 0.75 point

Sample 1

- $V@R = 80^{\text{th}}$ percentile = 180%
- $TV@R = (180 + 200)/2 = 190\%$
- $XTV@R = TV@R - \text{mean} = 190\% - \frac{(80+90+130+180+200)}{5} = 54\%$

Sample 2

- $VaR = E(X | X = 80\%) = 180\%$
- $TVaR = E(X | X > 80\%) = 200\%$
- $XTVaR = TVaR - \text{mean} = 200\% - 136\% = 64\%$

Sample 3

- $V@R = 180\%$
- $TV@R = \text{Area under curve where CDF} > 80\% = 180\%$
- $XTV@R = TV@R - \text{mean} = 180\% - 136\% = 44\%$

Part d: 0.25 point

Sample Responses

- WTVaR is TVaR calculated with probability distribution weighted to emphasize large loss.
- WTVaR is similar to TVaR, however the probabilities assigned to the simulations are assigned through a Wang Transformation instead of being uniform like TVaR.

Part e: 0.5 point

Sample Responses

- TVaR treats all losses equally. This doesn't always reflect true risk appetite. WTVaR corrects this by placing more weight on extreme values.
- TVaR treats losses linearly in the tail (i.e., a loss 2x as large is 2x as bad), which does not accurately reflect market attitudes. WTVaR is not linear and places higher weights on more extreme losses, which is more appropriate.

EXAMINER'S REPORT

Candidates were expected to compare and select a reinsurance program and provide statistics based off of a small set of simulations.

Part a

Candidates were expected to provide a plot of Combined Ratios vs CDF with the three sets of simulations plotted.

Common mistakes included:

- Providing alternative charts such as a Program vs Combined Ratio plot
- Plotting simulations on an axis
- Not sorting the observations
- Only providing discrete points for CDF

Part b

Candidates were expected to identify the proposed reinsurance program that would satisfy the actuary's concern about tail risk.

SAMPLE ANSWERS AND EXAMINER'S REPORT – SPRING 2019 EXAM 7

Common mistakes included: • Not selecting the program based on the concern over tail risk
Part c
Candidates were expected to provide a numerical solution for each of the three statistics using the direct data. There were several acceptable answers for TV@R and they were all given full credit. Common mistakes included: • V@R set at the 3rd observation (60th percent) rather than the 4th observation • Incorrect formula used for XTV@R • Calculating statistics for the current or proposed program instead of the direct book
Part d
Candidates were expected to describe that WTVaR is the same calculation as TVaR with transformed probabilities to put additional weight on the tail values. Common mistakes included: • Responding with only the acronym fully spelled out.
Part e
Candidates were expected to describe that TVaR does not properly reflect the risk averse attitude towards risk and that WTVaR corrects for this by applying additional weight on larger losses to reflect their increasing impact to the company. Common mistakes included: • Not stating the risk averseness or market attitude of increasingly larger losses.